

Coverage-Oriented Planner-Guided Memory Residual Reinforcement Learning for Robotic Thermal-Spot Coverage

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Abstract—Dynamic thermal processing creates spatially distributed thermal spots that remain active until a robot covers them. This paper studies coverage-oriented online service under changing target density, where a controller should improve coverage without discarding a reliable planner prior. We propose a planner-guided memory residual reinforcement learning pipeline: a Horizon-2 receding-horizon planner provides the base pursuit action, and an LSTM-PPO policy learns a bounded residual from active-spot, history, thermal, and route features. In held-out four-stage evaluation, the method matches Horizon-2 in low-density stages and improves coverage in the bursty Extreme stage, reaching 0.855 versus 0.816. It also avoids the collapse of direct learned controllers: vanilla LSTM-PPO and the no-residual variant fall to 0.061 and 0.687 in Extreme. A representative trajectory further shows a shorter path and fewer selected-target switches than Horizon-2. The results support planner-guided residual learning as a compact, coverage-oriented pipeline with an explicit coverage–service trade-off.

Index Terms—robotic coverage, residual reinforcement learning, coverage path planning, persistent monitoring, LSTM-PPO, MuJoCo

I. INTRODUCTION

Robotic thermal service differs from static coverage path planning: targets appear online, remain active until covered, and may become clustered under bursty operating conditions. Classical coverage planning and coverage control provide useful geometric structure [1], [2], [3], while persistent monitoring motivates measuring both coverage and response behavior [4]. Recent reinforcement-learning methods improve adaptability in coverage and robotics tasks [5], [6], [7], [8], but a fully learned policy can spend capacity rediscovering a competent planner and can fail when target density changes.

This paper keeps the planner inside the action loop. Inspired by residual robot learning [9], [10], [11], the learned policy only corrects a receding-horizon base action. The design is intentionally narrow: it is not a universal planner replacement,

but a planner-preserving pipeline for online thermal-spot coverage. The main contributions are: (i) a coverage-oriented dynamic thermal-spot service formulation; (ii) a memory residual PPO controller using active-spot attention and bounded action fusion; and (iii) a held-out evaluation showing clear advantages over direct learning baselines and the strongest Horizon-2 separation in the most bursty stage.

II. METHOD

Let $\mathcal{S}_t = \{(x_i, y_i, a_i)\}_{i=1}^{N_t}$ be the active thermal-spot set and $c_t = (x_t, y_t)$ the robot coverage center. A spot is removed when $\|c_t - (x_i, y_i)\|_2 \leq r_{\text{cover}}$. Coverage is the primary metric, $C = N_{\text{covered}}/N_{\text{spawned}}$; mean latency, p90 latency, and active backlog are reported as service diagnostics.

Fig. 1 summarizes the controller. Each active spot is encoded by relative position, distance, age, thermal/material context, reachability, and a risk score $q_i = w_a\phi_{\text{age}} + w_d\phi_{\text{dist}} + w_m\phi_{\text{mat}} + w_r\phi_{\text{reach}}$. A masked attention set encoder, following the permutation-invariant design pattern in Set Transformer [12], fuses up to eight active spots with robot state, spawn-history features, and route summaries. An LSTM state stores temporal context across decision steps.

The observation design reflects two properties of the task. First, the active set has no fixed order and its size changes as spots appear and disappear, so the policy should not depend on an arbitrary target index. The masked set encoder instead lets the policy compare the currently visible alternatives and ignore padded entries. Second, local geometry alone is not sufficient in a burst-lull process. Recent spawns, previous motion, and route summaries provide weak but useful evidence about whether the robot is repeatedly switching targets or moving through a compact service region.

The base action is $u_t^{\text{base}} = \pi_{\text{H2}}(s_t)$. The learned residual policy outputs $u_t^{\text{res}} = \pi_{\theta}(o_t, h_t)$ and executes

$$u_t = \text{clip}(u_t^{\text{base}} + \beta_t k_{\text{phase}} u_t^{\text{res}}), \quad (1)$$

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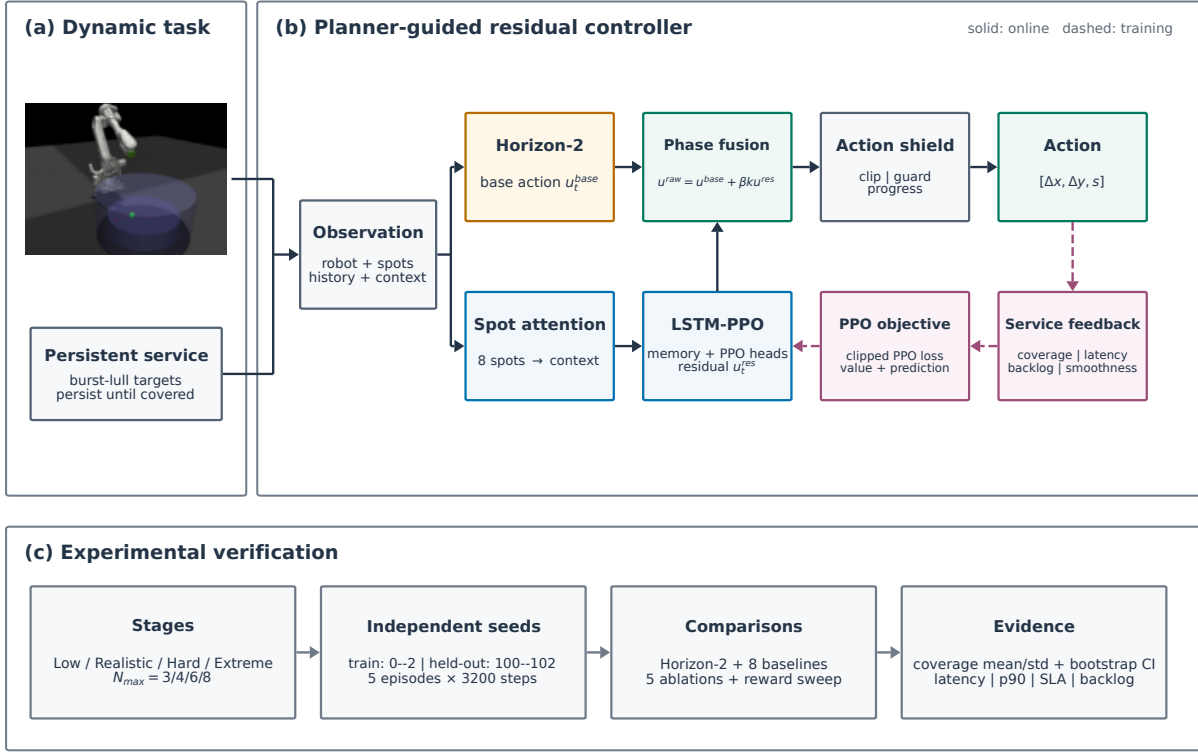


Fig. 1. Planner-guided residual PPO framework. A Horizon-2 planner supplies the base action, recurrent PPO predicts a bounded residual from active-spot and history features, and action fusion preserves the planner prior while allowing coverage-oriented correction under bursty target arrivals.

followed by direction and progress guards. The bounded scale and warm-up prevent early policy noise from overwhelming the planner. This is important in our task because a poor exploratory action can leave many thermal spots active for the rest of an episode. The residual branch is therefore used as a limited correction channel rather than as the main motion generator.

This fusion also clarifies the role of Horizon-2 in the pipeline. Horizon-2 is not treated only as an external baseline; it is the motion prior that defines a reasonable direction when the learned policy is uncertain. The residual can change that direction when memory indicates a better local trade-off, but the action remains clipped and aligned with the base motion. Thus, the comparison against Horizon-2 is intentionally strict: the learned component must improve the planner from inside the same control loop rather than replace it with a separate black-box policy.

The reward uses coverage as the dominant success term while retaining service pressure:

$$\begin{aligned}
 R_t = & \alpha_c \mathbb{I}[\text{covered}] + \alpha_f \psi(T_i) - \alpha_l \eta(T_i) \\
 & - \alpha_b N_t / N_{\max} - \alpha_o A_t^{\max} / A_{\text{norm}} \\
 & - \alpha_s \|u_t - u_{t-1}\|_2^2,
 \end{aligned} \quad (2)$$

where $\psi(T_i) = \max(0, 1 - T_i / T_{\text{SLA}})$ and $\eta(T_i) = \min(T_i / T_{\text{SLA}}, 2)$. The reward does not hide response-time costs; it makes coverage, latency, backlog, and smoothness part

of the same operating point. This is why the experiments report service diagnostics even though the primary target metric is coverage.

III. EXPERIMENTS AND RESULTS

Experiments use a MuJoCo-based robotic coverage simulator [13]. Four held-out stages increase maximum active targets from Low to Extreme. Learned policies are trained with seeds 0, 1, and 2 and evaluated on disjoint seeds 100, 101, and 102; each evaluation contains five 3200-step episodes. Horizon-2 is both the base planner and the primary comparison, while additional baselines test whether the result is specific to that planner. The evaluation is organized around three questions: whether coverage improves under bursty arrivals, whether the planner-residual interface prevents learned-policy collapse, and whether the resulting route is less locally reactive than the Horizon-2 baseline.

All reported metrics are computed from the same episode logs. Coverage counts covered spots over spawned spots, latency is measured from spot creation to successful coverage, and active backlog is averaged over the episode. The paper therefore avoids selecting a single favorable metric after evaluation. A method that covers more spots but leaves a larger queue is reported as such, and a method that is fast but misses many spots is not presented as coverage-equivalent. This accounting is especially important for the Extreme stage, where the coverage gain and the service cost appear together.

TABLE I
COVERAGE AND SERVICE SUMMARY AGAINST HORIZON-2

Stage	Ours Cov.	H2 Cov.	Δ	95% CI	Ours Lat.	H2 Lat.	Ours p90	H2 p90	Active gap
Low	0.917 $\pm$ 0.010	0.919 $\pm$ 0.004	-0.002	[-0.009,+0.006]	12.5s	12.2s	24.1s	23.7s	+0.02
Realistic	0.901 $\pm$ 0.016	0.896 $\pm$ 0.009	+0.005	[-0.009,+0.017]	16.0s	16.5s	29.7s	29.4s	+0.02
Hard	0.870 $\pm$ 0.024	0.870 $\pm$ 0.019	-0.000	[-0.024,+0.021]	19.6s	19.2s	38.1s	35.4s	+0.06
Extreme	0.855 $\pm$ 0.024	0.816 $\pm$ 0.026	+0.038	[+0.012,+0.068]	23.9s	21.4s	46.0s	39.9s	+0.26

Note: Ours aggregates three training seeds and three held-out evaluation seeds; Horizon-2 uses the held-out evaluation seeds. Active gap is Ours minus Horizon-2 average active backlog. The largest coverage separation appears in Extreme, with a visible latency/backlog cost.

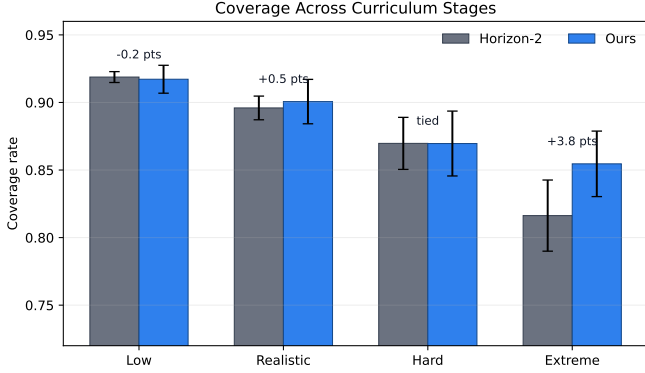


Fig. 2. Coverage comparison across held-out stages. The proposed method stays close to Horizon-2 in Low, Realistic, and Hard stages, then gives the clearest gain in Extreme, where target arrivals are densest. Error bars show variation across evaluation seeds.

Table I shows the main result. The residual policy is close to Horizon-2 in Low, Realistic, and Hard stages, but separates in Extreme by +0.038 coverage. This is the paper’s main advantage claim: memory residual learning helps most when the target field becomes bursty. The same row also shows the cost of that advantage: higher p90 latency and backlog in Extreme. We therefore present the method as a coverage-oriented operating point, not as a uniformly faster service policy.

Fig. 2 visualizes the same frontier in a form that is easier to read than the table alone. The advantage is not a uniform gain at every density level. Instead, it is concentrated where the planner has the hardest local-choice problem: the Extreme stage. This concentration is important for the paper’s claim because the method is designed to retain a competent short-horizon planner and add memory-based correction only when the active target field becomes difficult.

The pattern is consistent with the intended role of memory. In low-density operation, a short-horizon planner can already pursue visible targets effectively, so a learned correction has little room to improve. In the Extreme stage, active targets are more likely to accumulate and interact, and the recurrent residual can use recent spawn and route context to adjust the planner’s next action. The result is not a blanket improvement over every heuristic, but a more robust learned pipeline than direct PPO-style control.

This interpretation also explains why the Realistic and Hard

TABLE II
BASELINE AND DIAGNOSTIC HIGHLIGHTS

Method or diagnostic	Low	Realistic	Hard	Extreme
Ours	0.917	0.901	0.870	0.855
Horizon-2	0.919	0.896	0.870	0.816
Risk-aware heuristic	0.924	0.901	0.836	0.863
No residual PPO	0.838	0.776	0.726	0.687
Vanilla LSTM-PPO	0.166	0.130	0.089	0.061
Path length, Realistic	Ours 15.37		H2 16.40	
Target switches, Realistic	Ours 25		H2 40	

stages are kept in the table even when the numerical differences are small. They show that the residual branch does not buy Extreme-stage coverage by destroying normal operation. In those stages, the confidence intervals overlap with Horizon-2 and the service diagnostics remain close, which is useful evidence for a planner-guided design. The method is best read as a controlled extension of a strong planner: conservative where the planner is already sufficient and more active when dense target interactions make short-horizon choices harder.

Table II condenses the broader evidence. First, direct learned controllers are not competitive: removing the planner residual interface drops Extreme coverage from 0.855 to 0.687, and vanilla LSTM-PPO collapses to 0.061. This supports the pipeline design more strongly than any single auxiliary module. Auxiliary ablations were mixed: removing attention, carry, prediction, or service shaping sometimes changes a moderate stage by a few thousandths, but none changes the main conclusion as strongly as removing the planner residual interface. Second, the risk-aware heuristic remains strong and slightly exceeds our Extreme coverage, which bounds the claim and prevents overstatement. The proposed method’s advantage is that it combines planner-level stability with learned memory correction, rather than relying on a fully learned controller.

Fig. 3 adds route-level evidence for the same conclusion. In the representative Realistic episode, Horizon-2 repeatedly redirects toward locally attractive targets and forms a wider looping pattern. The residual policy keeps the planner prior but biases the motion toward a more compact service route. This behavior is consistent with the lower path length and fewer selected-target switches in Table II. The trajectory comparison therefore supports a practical advantage beyond final coverage:

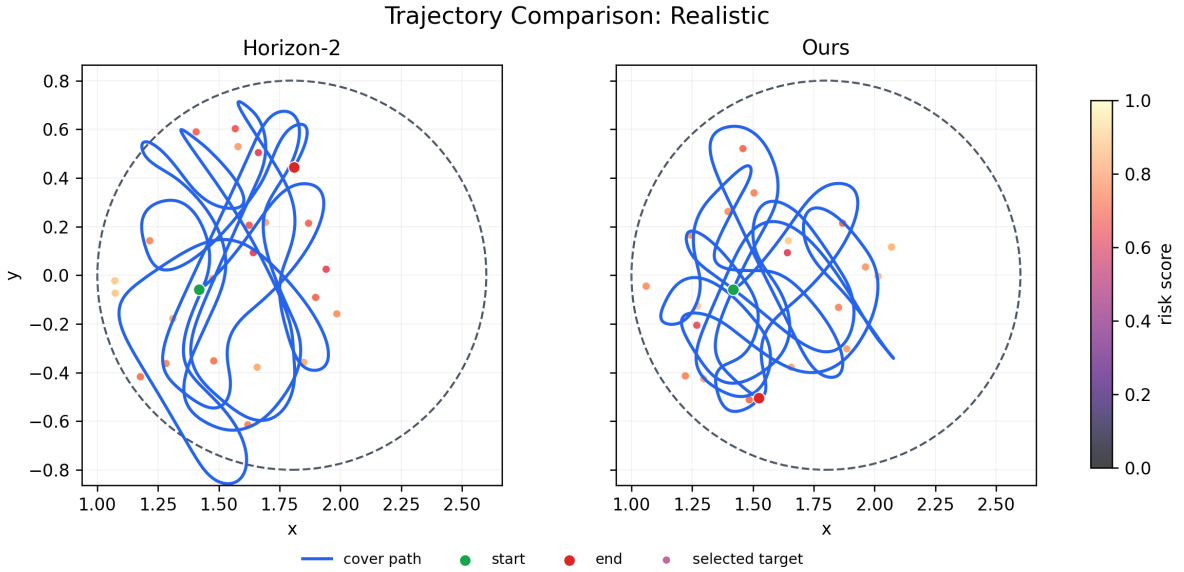


Fig. 3. Representative trajectory comparison in the Realistic stage. Compared with Horizon-2, planner-guided residual PPO follows a more compact route around the active thermal spots, with shorter path length (15.37 versus 16.40) and fewer selected-target switches (25 versus 40). The trajectory is used as a qualitative diagnostic and is interpreted together with the seed-averaged metrics in Tables I and II.

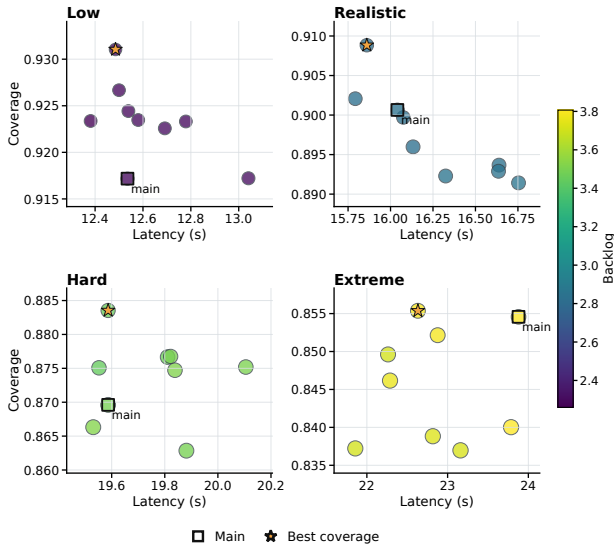


Fig. 4. Reward-weight sweep around the final policy setting. The sweep shows that nearby weights move the controller between coverage-oriented and service-oriented operating points, which explains why coverage gains should be read together with latency and backlog diagnostics.

the robot can spend less motion on repeated retargeting while maintaining high coverage. We treat this as a diagnostic trajectory rather than a replacement for aggregate evaluation, so the main performance claim remains anchored in the held-out seed averages.

The reward-sweep experiments in Fig. 4 further support the same interpretation: nearby reward weights move the policy between coverage-oriented and service-oriented behavior. This

matters for industrial use because an operator may prefer the highest coverage under bursty arrivals, while another deployment may require lower backlog or p90 latency. In this conference version, Table I keeps the trade-off visible by reporting coverage, mean latency, p90 latency, and backlog in the same row. The table, coverage figure, trade-off sweep, and trajectory figure together separate three forms of evidence: stronger burst-stage coverage, a controllable service frontier, and a less reactive representative route.

IV. DISCUSSION AND CONCLUSION

The evidence supports three focused conclusions. First, planner-guided residual learning is substantially safer than direct PPO in this task: both no-residual and vanilla LSTM-PPO fail under high target density. Second, the learned residual has its clearest practical value in bursty operation, where history and route context help the controller move beyond myopic target pursuit. Third, coverage and response time are not the same objective. In Extreme, higher coverage comes with higher latency and backlog, so deployment should choose the reward setting according to whether coverage or response time is more important.

The figures clarify these conclusions from complementary views. Fig. 2 gives the main quantitative picture: the proposed method is not intended to dominate a strong planner in every easy stage, but to improve the hardest bursty stage while remaining close elsewhere. Fig. 3 explains why the residual branch can be useful even when aggregate coverage values are close. Horizon-2 repeatedly follows the currently attractive target and can create wider loops; the memory residual policy keeps the planner’s direction but dampens some repeated retargeting. This route-level behavior is consistent with the lower

path length and fewer selected-target switches in Table II. Fig. 4 then shows the boundary of the advantage: the final policy is a coverage-oriented point on a frontier, not a latency-minimizing point.

For practical use, this distinction matters. A thermal-service system may prefer the residual controller when missed coverage under dense arrivals is more costly than a moderate increase in response time. A latency-critical deployment may instead select a different reward weight or a conservative heuristic. The main contribution is therefore a planner-guided learning interface that can exploit memory without abandoning a stable planner prior. This makes the method easier to interpret than direct PPO control and less brittle than a learned policy that must generate the whole action from scratch.

The current experiments also indicate where the method should not be overclaimed. The risk-aware heuristic remains competitive, and the residual policy is not uniformly better than Horizon-2 on every diagnostic. These results suggest that the main value is not a universal dominance claim, but an architecture choice: using learning to adjust planner behavior under history-dependent congestion. This is a useful operating mode for thermal coverage because the physical task often penalizes missed or repeatedly deferred regions, while still requiring a controller whose behavior can be traced back to a conventional planner.

The evaluation protocol was also chosen to make this distinction reproducible. Training and evaluation use disjoint seeds, and the main comparisons keep the same stage definitions and episode length across methods. The reported diagnostics therefore separate three decisions that are often mixed together in online service papers: the motion prior supplied by the planner, the residual correction learned from history, and the reward weight that selects a coverage-service operating point. This separation should make it easier to reproduce the comparison or replace Horizon-2 with another receding-horizon planner in future work.

This paper therefore positions planner-guided memory residual PPO as a compact coverage-oriented pipeline for dynamic thermal-spot service. The method preserves a strong Horizon-2 prior, improves coverage in the most bursty setting, and produces a smoother representative route with fewer target switches. The current version keeps the framework figure, the core quantitative tables, and one trajectory comparison so that the main algorithmic route and experimental advantages are both visible before the references. Future work should add explicit latency constraints or adaptive reward selection so that the coverage-service operating point can be selected automatically for a given industrial requirement.

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